

Illustrated calculus certificates

23 typogeometric OEIS cases

This portfolio presents one concise certificate per case: defining relation, first terms, typogeometric trees, exact contour extraction, recurrence, differential equation, and matrix-reduction dimensions. Each constituent PDF embeds its exact JSON payload; this combined PDF embeds a SHA-256 index pointing to every constituent payload and certificate.

Scope. A120588–A120607, A244594, A244627, and A244856.

Quality status. All 23 defining relations, published prefixes, recurrence instances, PDF structures, embedded payloads, and page layouts pass the release audit.

Detailed reference. The reviewed A120590 q=3 paper remains separately available as the verbose explanatory example; these pages deliberately filter each record to its most useful verified facts.

Definition and first terms

$3\,A(x)=2+1\,x+A(x)^2$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+1*T^2$.
 $a(0),\ldots,a(7)=1,1,1,2,5,14,42,132$.
 Kernel used throughout: $\rho(u)=u\,(1-1\,u^1)$, so $x=\rho(T(x))$.

Typogeometry

1x1

{1,1}x1

{1,1}x1

{{1,1},1}x1

literal 2-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2\colon 1$.

Coefficient and contour extraction

$$\mathcal{K}_n=\{(k_1,\ldots,k_1)\in\mathbb{Z}_{\geq 0}^1:\sum_{j=1}^1jk_j=n-1\},\quad K=\sum_{j=1}^1k_j,$$

$$a(n)=\frac{1}{n}\sum_{\mathbf{k}\in\mathcal{K}_n}\binom{n+K-1}{K}\binom{K}{k_1,\ldots,k_1}1^{k_1}.$$

$$a(n)=\frac{1}{2\pi i n}\oint_{\gamma}\frac{du}{\rho(u)^n},\quad n\geq 1.$$

Recurrence

$$\sum_{r=0}^1P_r(n)a(n+r)=0,\,n\geq 1,\,\text{with}$$

$$P_0(n)=-4\,n+2$$

$$P_1(n)=n+1$$

Differential equation

$$(2\,x)\,A(x)+(-4\,x^2+x)\,A'(x)=3\,x.$$

Matrices, telescope certificate, and checks

$$G=\begin{pmatrix}0&0&-1&0\\1&0&2&-1\\-1&1&0&2\\0&-1&0&0\end{pmatrix},\quad U=\begin{pmatrix}4&2\\0&0\end{pmatrix},\quad V=\begin{pmatrix}-1&0\\2&1\end{pmatrix},\quad J=\begin{pmatrix}0&1\\0&0\end{pmatrix}$$

Certificate.

$$R(n,u)=\frac{2\,u-1}{\rho(u)^0}.$$

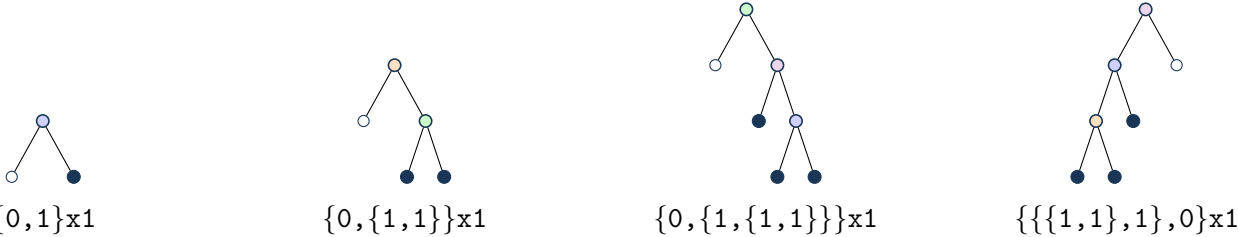
Telescoping identity: $\sum_rP_r(n)H_{n+r}(u)=\partial_u(R(n,u)H_n(u))$.

Matrix dimensions: 4×4 ; remainder 1×2 , rank 1, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$A(x) = A_{\text{parent}}(x)^2$, $A_{\text{parent}}(x) = A_{120588}(x)$, with the origin-normalized solution.
Normalization/grammar: ordered forest of 2 parent A120588 typogeometries.
 $a(0), \dots, a(7) = 1, 2, 3, 6, 15, 42, 126, 396$.
Kernel used throughout: $\rho(u) = u(1 - 1u^1)$, so $x = \rho(T(x))$.

Typogeometry



forest slots are explicit; 0 denotes an empty slot. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.
Typogeometric constructor codes and multiplicities: Δ_2 : 1.

Coefficient and contour extraction

$$p(n) = [x^n] A_{\text{parent}}(x),$$
$$a(n) = \sum_{\substack{n_1, \dots, n_2 \geq 0 \\ n_1 + \dots + n_2 = n}} \prod_{j=1}^2 p(n_j).$$

$$[x^n] A(x)^2 = \frac{2}{2\pi i n} \oint_{\gamma} \frac{(1+u)^1 du}{u^n D(u)^n}.$$

Recurrence

$$\sum_{r=0}^2 P_r(n) a(n+r) = 0, \quad n \geq 1, \text{ with}$$
$$P_0(n) = 0$$
$$P_1(n) = -4n - 2$$
$$P_2(n) = n + 2$$

Differential equation

$$(2x) A(x) + (-4x^2 + x) A'(x) = 2x^2 + 4x.$$

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.
Certificate.

$$R(n, u) = N(n, u)/(-u^2 + u)^1.$$

$$\text{Telescoping identity: } \sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u)).$$

Matrix dimensions: 4×4 ; remainder 2×3 , rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

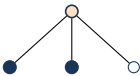
Definition and first terms

$4\,A(x)=3+1\,x+A(x)^3$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+3*T^2+1*T^3$.
 $a(0),\ldots,a(7)=1,1,3,19,150,1326,12558,124590$.
 Kernel used throughout: $\rho(u)=u\,(1-3\,u^1-1\,u^2)$, so $x=\rho(T(x))$.

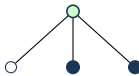
Typogeometry



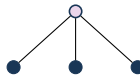
1x1



{1,1,0}x1



{0,1,1}x1



{1,1,1}x1

literal 3-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2\colon 3$, $\Delta_3\colon 1$.

Coefficient and contour extraction

$$\mathcal{K}_n=\{(k_1,\ldots,k_2)\in\mathbb{Z}_{\geq 0}^2:\sum_{j=1}^2jk_j=n-1\},\quad K=\sum_{j=1}^2k_j,$$

$$a(n)=\frac{1}{n}\sum_{\mathbf{k}\in\mathcal{K}_n}\binom{n+K-1}{K}\binom{K}{k_1,\ldots,k_2}3^{k_1}1^{k_2}.$$

$$a(n)=\frac{1}{2\pi i n}\oint_\gamma\frac{du}{\rho(u)^n},\quad n\geq 1.$$

Recurrence

$$\sum_{r=0}^2P_r(n)a(n+r)=0,\,n\geq 1,\,\text{with}$$

$$\begin{aligned}P_0(n)&=-27\,n^2+3\\P_1(n)&=-162\,n^2-243\,n-81\\P_2(n)&=13\,n^2+39\,n+26\end{aligned}$$

Differential equation

$$(3\,x^2)\,A(x)+(-27\,x^3-81\,x^2)\,A'(x)+(-27\,x^4-162\,x^3+13\,x^2)\,A^{(2)}(x)=0.$$

Matrices, telescoper certificate, and checks

$$U=\begin{pmatrix}\frac{153}{13}&\frac{24}{13}&\frac{3}{13}\\\frac{72}{13}&\frac{9}{13}&\frac{6}{13}\\0&0&0\end{pmatrix},\quad V=\begin{pmatrix}-1&0&0\\\frac{75}{13}&\frac{11}{13}&\frac{3}{13}\\\frac{24}{13}&\frac{3}{13}&\frac{2}{13}\end{pmatrix},\quad J=\begin{pmatrix}0&1&0\\0&0&2\\0&0&0\end{pmatrix}$$

Certificate.

$$R(n,u)=\frac{-9\,n\,u^4-36\,n\,u^3+6\,n\,u^2+84\,n\,u-13\,n-3\,u^4-12\,u^3-6\,u^2+3\,u}{\rho(u)^1}.$$

$$\text{Telescoping identity: }\sum_rP_r(n)H_{n+r}(u)=\partial_u(R(n,u)H_n(u)).$$

Matrix dimensions: 6×6 ; remainder 2×3 , rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

A120591 concise calculus certificate

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VERIFIED

July 31, 2026

Definition and first terms

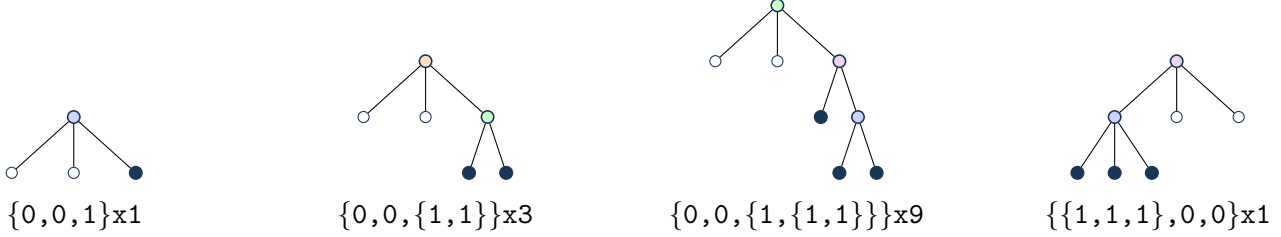
$A(x) = A_{\text{parent}}(x)^3$, $A_{\text{parent}}(x) = A_{120590}(x)$, with the origin-normalized solution.

Normalization/grammar: **ordered forest of 3 parent A120590 typogeometries**.

$a(0), \dots, a(7) = 1, 3, 12, 76, 600, 5304, 50232, 498360$.

Kernel used throughout: $\rho(u) = u(1 - 3u^1 - 1u^2)$, so $x = \rho(T(x))$.

Typogeometry



forest slots are explicit; 0 denotes an empty slot. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2: 3$, $\Delta_3: 1$.

Coefficient and contour extraction

$$p(n) = [x^n] A_{\text{parent}}(x),$$

$$a(n) = \sum_{\substack{n_1, \dots, n_3 \geq 0 \\ n_1 + \dots + n_3 = n}} \prod_{j=1}^3 p(n_j).$$

$$[x^n] A(x)^3 = \frac{3}{2\pi i n} \oint_{\gamma} \frac{(1+u)^2 du}{u^n D(u)^n}.$$

Recurrence

$\sum_{r=0}^3 P_r(n) a(n+r) = 0$, $n \geq 1$, with

$$P_0(n) = 0$$

$$P_1(n) = -27n^2 - 54n - 24$$

$$P_2(n) = -162n^2 - 567n - 486$$

$$P_3(n) = 13n^2 + 65n + 78$$

Differential equation

$$(3x^2) A(x) + (-27x^3 - 81x^2) A'(x) + (-27x^4 - 162x^3 + 13x^2) A^{(2)}(x) = 24x^3 + 72x^2.$$

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n, u) = N(n, u) / (-u^3 - 3u^2 + u)^2.$$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 6×6 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$5\,A(x) = 4 + 4\,x + A(x)^3$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(2)*T(x)$; $T=x+3*T^2+2*T^3$.
 $a(0),\ldots,a(7) = 1, 2, 6, 40, 330, 3048, 30156, 312528$.
 Kernel used throughout: $\rho(u) = u\,(1-3\,u^1-2\,u^2)$, so $x = \rho(T(x))$.

Typogeometry

1×2
 $\{1, 1\} \times 6$
 $\{1, \{1, 1\}\} \times 18$
 $\{1, 1, 1\} \times 4$

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 3, Δ_3 : 2.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_2) \in \mathbb{Z}_{\geq 0}^2 : \sum_{j=1}^2 j k_j = n - 1\}, \quad K = \sum_{j=1}^2 k_j,$$

$$a(n) = \frac{2}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \ldots, k_2} 3^{k_1} 2^{k_2}.$$

$$a(n) = \frac{2}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^2 P_r(n) a(n+r) = 0, \, n \geq 1, \text{ with}$

$P_0(n) = -108\,n^2 + 12$
 $P_1(n) = -216\,n^2 - 324\,n - 108$
 $P_2(n) = 17\,n^2 + 51\,n + 34$

Differential equation

$$(12\,x^2)\,A(x) + (-108\,x^3 - 108\,x^2)\,A'(x) + (-108\,x^4 - 216\,x^3 + 17\,x^2)\,A^{(2)}(x) = 0.$$

Matrices, telescoper certificate, and checks

$$U = \begin{pmatrix} \frac{198}{17} & \frac{30}{17} & \frac{3}{17} \\ \frac{180}{17} & \frac{18}{17} & \frac{12}{17} \\ 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 & 0 \\ \frac{96}{17} & \frac{13}{17} & \frac{3}{17} \\ \frac{60}{17} & \frac{6}{17} & \frac{4}{17} \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

Certificate.

$$R(n,u) = \frac{-72\,n\,u^4 - 144\,n\,u^3 + 42\,n\,u^2 + 114\,n\,u - 17\,n - 24\,u^4 - 48\,u^3 - 6\,u^2 + 6\,u}{\rho(u)^1}.$$

$$\text{Telescoping identity: } \sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n,u) H_n(u)).$$

Matrix dimensions: 6×6 ; remainder 2×3 , rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$5\,A(x) = 4 + 1\,x + A(x)^4$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+6*T^2+4*T^3+1*T^4$.
 $a(0),\dots,a(7) = 1, 1, 6, 76, 1201, 21252, 402892, 8001412$.
 Kernel used throughout: $\rho(u) = u\,(1 - 6\,u^1 - 4\,u^2 - 1\,u^3)$, so $x = \rho(T(x))$.

Typogeometry

literal 4-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 6, Δ_3 : 4, Δ_4 : 1.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_3) \in \mathbb{Z}_{\geq 0}^3 : \sum_{j=1}^3 j k_j = n - 1\}, \quad K = \sum_{j=1}^3 k_j,$$

$$a(n) = \frac{1}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_3} 6^{k_1} 4^{k_2} 1^{k_3}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^3 P_r(n) a(n+r) = 0, \, n \geq 1, \text{ with}$$

$$P_0(n) = -256\,n^3 - 384\,n^2 - 48\,n + 40$$

$$P_1(n) = -3072\,n^3 - 9216\,n^2 - 8896\,n - 2752$$

$$P_2(n) = -12288\,n^3 - 55296\,n^2 - 79872\,n - 36864$$

$$P_3(n) = 491\,n^3 + 2946\,n^2 + 5401\,n + 2946$$

Differential equation

$$\begin{aligned} & (40\,x^3)\,A(x) + (-688\,x^4 - 2752\,x^3)\,A'(x) \\ & + (-1152\,x^5 - 9216\,x^4 - 18432\,x^3)\,A^{(2)}(x) + (-256\,x^6 - 3072\,x^5 - 12288\,x^4 + 491\,x^3)\,A^{(3)}(x) = 0. \end{aligned}$$

Matrices, telescoper certificate, and checks

$$U = \begin{pmatrix} \frac{11536}{491} & \frac{900}{491} & \frac{60}{491} & \frac{4}{491} \\ \frac{11040}{491} & \frac{736}{491} & \frac{180}{491} & \frac{12}{491} \\ \frac{3600}{491} & \frac{240}{491} & \frac{16}{491} & \frac{132}{491} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 & 0 & 0 \\ \frac{5644}{491} & \frac{409}{491} & \frac{60}{491} & \frac{4}{491} \\ \frac{3660}{491} & \frac{244}{491} & \frac{49}{491} & \frac{36}{491} \\ \frac{900}{491} & \frac{60}{491} & \frac{4}{491} & \frac{33}{491} \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

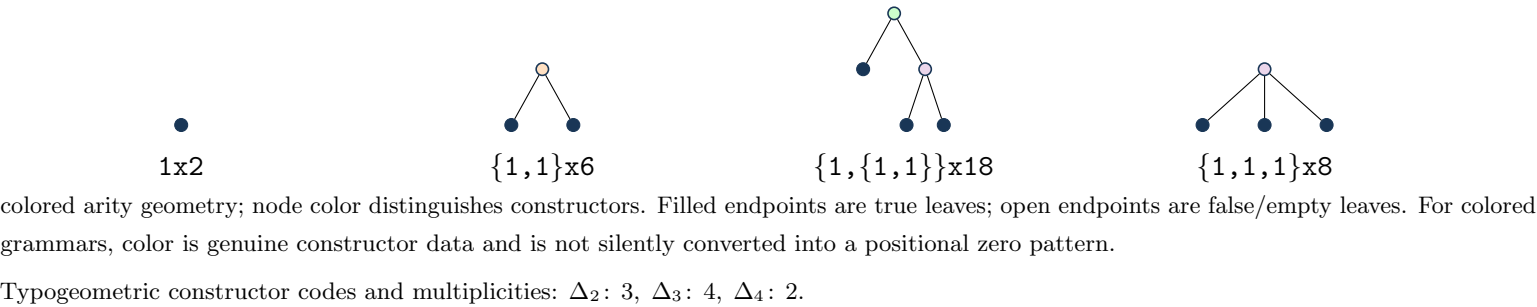
Certificate.

$R(n,u) = N(n,u)/\rho(u)^2$, $\deg_u N = 9$.
 Telescoping identity: $\sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u))$.
 Matrix dimensions: 8×8 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$8\,A(x)=7+8\,x+A(x)^4$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(2)*T(x)$; $T=x+3*T^2+4*T^3+2*T^4$.
 $a(0),\ldots,a(7)=1,2,6,44,394,3948,42364,476120$.
 Kernel used throughout: $\rho(u)=u\,(1-3\,u^1-4\,u^2-2\,u^3)$, so $x=\rho(T(x))$.

Typogeometry



Coefficient and contour extraction

$$\mathcal{K}_n=\{(k_1,\ldots,k_3)\in\mathbb{Z}_{\geq 0}^3:\sum_{j=1}^3jk_j=n-1\},\quad K=\sum_{j=1}^3k_j,$$

$$a(n)=\frac{2}{n}\sum_{\mathbf{k}\in\mathcal{K}_n}\binom{n+K-1}{K}\binom{K}{k_1,\ldots,k_3}3^{k_1}4^{k_2}2^{k_3}.$$

$$a(n)=\frac{2}{2\pi i n}\oint_{\gamma}\frac{du}{\rho(u)^n},\quad n\geq 1.$$

Recurrence

$$\sum_{r=0}^3P_r(n)a(n+r)=0,\,n\geq 1,\,\text{with}$$

$$\begin{aligned}P_0(n)&=-512\,n^3-768\,n^2-96\,n+80\\P_1(n)&=-1344\,n^3-4032\,n^2-3892\,n-1204\\P_2(n)&=-1176\,n^3-5292\,n^2-7644\,n-3528\\P_3(n)&=89\,n^3+534\,n^2+979\,n+534\end{aligned}$$

Differential equation

$$\begin{aligned}(80\,x^3)\,A(x)+(-1376\,x^4-1204\,x^3)\,A'(x)\\+(-2304\,x^5-4032\,x^4-1764\,x^3)\,A^{(2)}(x)+(-512\,x^6-1344\,x^5-1176\,x^4+89\,x^3)\,A^{(3)}(x)=0.\end{aligned}$$

Matrices, telescoper certificate, and checks

$$U=\begin{pmatrix}\frac{1016}{89}&\frac{144}{89}&\frac{12}{89}&\frac{1}{89}\\ \frac{1824}{89}&\frac{152}{89}&\frac{72}{89}&\frac{6}{89}\\ \frac{1152}{89}&\frac{96}{89}&\frac{8}{89}&\frac{60}{89}\\ 0&0&0&0\end{pmatrix},\quad V=\begin{pmatrix}-1&0&0&0\\ \frac{482}{89}&\frac{55}{89}&\frac{12}{89}&\frac{1}{89}\\ \frac{600}{89}&\frac{50}{89}&\frac{19}{89}&\frac{9}{89}\\ \frac{288}{89}&\frac{24}{89}&\frac{2}{89}&\frac{15}{89}\end{pmatrix},\quad J=\begin{pmatrix}0&1&0&0\\ 0&0&2&0\\ 0&0&0&3\\ 0&0&0&0\end{pmatrix}$$

Certificate.

$$R(n,u)=N(n,u)/\rho(u)^2,\,\deg_u N=9.$$

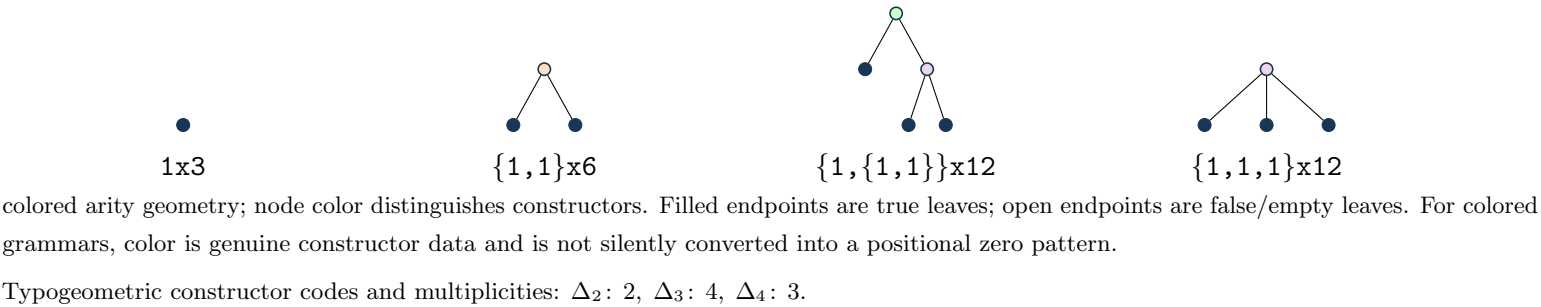
$$\text{Telescoping identity: }\sum_rP_r(n)H_{n+r}(u)=\partial_u(R(n,u)H_n(u)).$$

Matrix dimensions: 8×8 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$13 A(x) = 12 + 27 x + A(x)^4$, with the origin-normalized solution.
Normalization/grammar: $A(x)=1+(3)*T(x)$; $T=x+2*T^2+4*T^3+3*T^4$.
 $a(0), \dots, a(7) = 1, 3, 6, 36, 249, 1932, 16044, 139500$.
Kernel used throughout: $\rho(u) = u \left(1 - 2 u^1 - 4 u^2 - 3 u^3\right)$, so $x = \rho(T(x))$.

Typogeometry



Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_3) \in \mathbb{Z}_{\geq 0}^3 : \sum_{j=1}^3 j k_j = n - 1\}, \quad K = \sum_{j=1}^3 k_j,$$
$$a(n) = \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_3} 2^{k_1} 4^{k_2} 3^{k_3}.$$
$$a(n) = \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^3 P_r(n) a(n+r) = 0, \quad n \geq 1, \text{ with}$$
$$\begin{aligned} P_0(n) &= -6912 n^3 - 10368 n^2 - 1296 n + 1080 \\ P_1(n) &= -9216 n^3 - 27648 n^2 - 26688 n - 8256 \\ P_2(n) &= -4096 n^3 - 18432 n^2 - 26624 n - 12288 \\ P_3(n) &= 451 n^3 + 2706 n^2 + 4961 n + 2706 \end{aligned}$$

Differential equation

$$\begin{aligned} &(1080 x^3) A(x) + (-18576 x^4 - 8256 x^3) A'(x) \\ &+ (-31104 x^5 - 27648 x^4 - 6144 x^3) A^{(2)}(x) + (-6912 x^6 - 9216 x^5 - 4096 x^4 + 451 x^3) A^{(3)}(x) = 0. \end{aligned}$$

Matrices, telescoper certificate, and checks

$$U = \begin{pmatrix} \frac{3376}{451} & \frac{676}{451} & \frac{52}{451} & \frac{4}{451} \\ \frac{8736}{451} & \frac{672}{451} & \frac{468}{451} & \frac{36}{451} \\ \frac{8112}{451} & \frac{624}{451} & \frac{48}{451} & \frac{420}{451} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 & 0 & 0 \\ \frac{1572}{451} & \frac{225}{451} & \frac{52}{451} & \frac{4}{451} \\ \frac{260}{41} & \frac{20}{41} & \frac{11}{41} & \frac{4}{41} \\ \frac{2028}{451} & \frac{156}{451} & \frac{12}{451} & \frac{105}{451} \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Certificate.

$$R(n, u) = N(n, u)/\rho(u)^2, \quad \deg_u N = 9.$$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 8×8 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

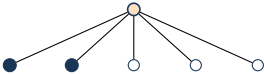
Definition and first terms

$6 A(x) = 5 + 1 x + A(x)^5$, with the origin-normalized solution.
Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+10*T^2+10*T^3+5*T^4+1*T^5$.
 $a(0), \dots, a(7) = 1, 1, 10, 210, 5505, 161601, 5082420, 167451780$.
Kernel used throughout: $\rho(u) = u(1 - 10 u^1 - 10 u^2 - 5 u^3 - 1 u^4)$, so $x = \rho(T(x))$.

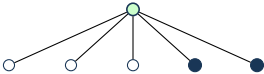
Typogeometry



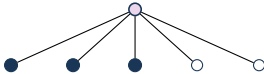
1x1



{1,1,0,0,0}x1



{0,0,0,1,1}x1



{1,1,1,0,0}x1

literal 5-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 10, Δ_3 : 10, Δ_4 : 5, Δ_5 : 1.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_4) \in \mathbb{Z}_{\geq 0}^4 : \sum_{j=1}^4 j k_j = n - 1\}, \quad K = \sum_{j=1}^4 k_j,$$
$$a(n) = \frac{1}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_4} 10^{k_1} 10^{k_2} 5^{k_3} 1^{k_4}.$$
$$a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^4 P_r(n) a(n+r) = 0, \ n \geq 1$, with

$$\begin{aligned} P_0(n) &= -3125 n^4 - 12500 n^3 - 13750 n^2 - 2500 n + 1155 \\ P_1(n) &= -62500 n^4 - 343750 n^3 - 668750 n^2 - 546875 n - 159375 \\ P_2(n) &= -468750 n^4 - 3281250 n^3 - 8390625 n^2 - 9234375 n - 3656250 \\ P_3(n) &= -1562500 n^4 - 13281250 n^3 - 40625000 n^2 - 52343750 n - 23437500 \\ P_4(n) &= 37531 n^4 + 375310 n^3 + 1313585 n^2 + 1876550 n + 900744 \end{aligned}$$

Differential equation

The scalar linear ODE has order 4. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `case.json#/objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u)/\rho(u)^3, \ \deg_u N = 16.$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u)).$

Matrix dimensions: 10×10 ; remainder 4×5 , rank 4, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$9\,A(x) = 8 + 8\,x + A(x)^5$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(2)*T(x)$; $T=x+5*T^2+10*T^3+10*T^4+4*T^5$.
 $a(0),\ldots,a(7) = 1, 2, 10, 120, 1770, 29208, 516180, 9554640$.
 Kernel used throughout: $\rho(u) = u\,(1 - 5\,u^1 - 10\,u^2 - 10\,u^3 - 4\,u^4)$, so $x = \rho(T(x))$.

Typogeometry

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 5, Δ_3 : 10, Δ_4 : 10, Δ_5 : 4.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_4) \in \mathbb{Z}_{\geq 0}^4 : \sum_{j=1}^4 j k_j = n - 1\}, \quad K = \sum_{j=1}^4 k_j,$$

$$a(n) = \frac{2}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \ldots, k_4} 5^{k_1} 10^{k_2} 10^{k_3} 4^{k_4}.$$

$$a(n) = \frac{2}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^4 P_r(n) a(n+r) = 0, \, n \geq 1, \text{ with}$$

$$\begin{aligned} P_0(n) &= -50000\,n^4 - 200000\,n^3 - 220000\,n^2 - 40000\,n + 18480 \\ P_1(n) &= -200000\,n^4 - 1100000\,n^3 - 2140000\,n^2 - 1750000\,n - 510000 \\ P_2(n) &= -300000\,n^4 - 2100000\,n^3 - 5370000\,n^2 - 5910000\,n - 2340000 \\ P_3(n) &= -200000\,n^4 - 1700000\,n^3 - 5200000\,n^2 - 6700000\,n - 3000000 \\ P_4(n) &= 9049\,n^4 + 90490\,n^3 + 316715\,n^2 + 452450\,n + 217176 \end{aligned}$$

Differential equation

The scalar linear ODE has order 4. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#/objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n,u) = N(n,u)/\rho(u)^3$, $\deg_u N = 16$.
 Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n,u) H_n(u))$.
 Matrix dimensions: 10×10 ; remainder 4×5 , rank 4, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$30\,A(x) = 29 + 125\,x + A(x)^5$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(5)*T(x)$; $T=x+2*T^2+10*T^3+25*T^4+25*T^5$.
 $a(0), \dots, a(7) = 1, 5, 10, 90, 825, 8445, 92820, 1066740$.
 Kernel used throughout: $\rho(u) = u\,(1 - 2\,u^1 - 10\,u^2 - 25\,u^3 - 25\,u^4)$, so $x = \rho(T(x))$.

Typogeometry

1x5

{1,1}x10

{1,{1,1}}x20

{1,1,1}x50

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 2, Δ_3 : 10, Δ_4 : 25, Δ_5 : 25.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_4) \in \mathbb{Z}_{\geq 0}^4 : \sum_{j=1}^4 j k_j = n - 1\}, \quad K = \sum_{j=1}^4 k_j,$$

$$a(n) = \frac{5}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_4} 2^{k_1} 10^{k_2} 25^{k_3} 25^{k_4}.$$

$$a(n) = \frac{5}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^4 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-4 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 4. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u) / \rho(u)^3$, $\deg_u N = 16$.
 Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.
 Matrix dimensions: 10×10 ; remainder 4×5 , rank 4, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$13 A(x) = 12 + 32\, x + A(x)^5$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(4)*T(x)$; $T=x+5*T^2+20*T^3+40*T^4+32*T^5$.
 $a(0), \dots, a(7) = 1, 4, 20, 280, 4660, 86728, 1727880, 36047280$.
 Kernel used throughout: $\rho(u) = u\,(1 - 5\,u^1 - 20\,u^2 - 40\,u^3 - 32\,u^4)$, so $x = \rho(T(x))$.

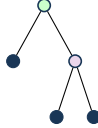
Typogeometry



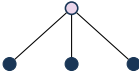
1x4



{1,1}x20



{1,{1,1}}x100



{1,1,1}x80

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 5, Δ_3 : 20, Δ_4 : 40, Δ_5 : 32.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_4) \in \mathbb{Z}_{\geq 0}^4 : \sum_{j=1}^4 j k_j = n - 1\}, \quad K = \sum_{j=1}^4 k_j,$$

$$a(n) = \frac{4}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1, \dots, k_4} 5^{k_1} 20^{k_2} 40^{k_3} 32^{k_4}.$$

$$a(n) = \frac{4}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^4 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-4 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 4. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u)/\rho(u)^3$, $\deg_u N = 16$.
 Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.
 Matrix dimensions: 10×10 ; remainder 4×5 , rank 4, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

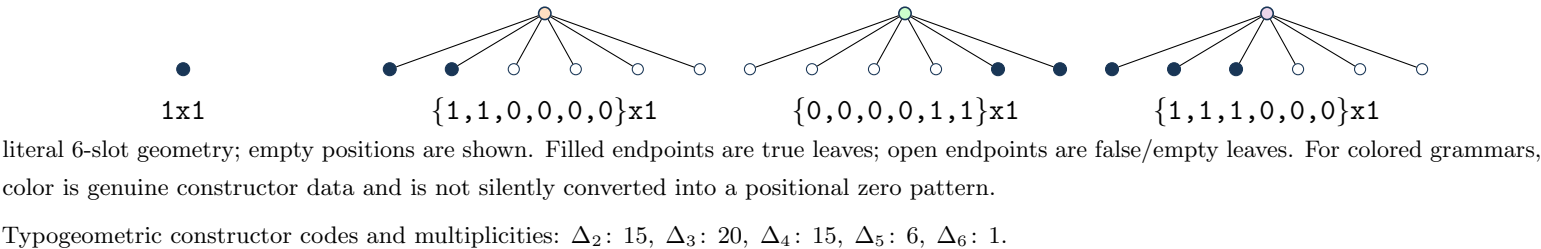
$7 A(x) = 6 + 1 x + A(x)^6$, with the origin-normalized solution.

Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+15*T^2+20*T^3+15*T^4+6*T^5+1*T^6$.

$a(0), \dots, a(7) = 1, 1, 15, 470, 18390, 805806, 37828981, 1860433080$.

Kernel used throughout: $\rho(u) = u (1 - 15 u^1 - 20 u^2 - 15 u^3 - 6 u^4 - 1 u^5)$, so $x = \rho(T(x))$.

Typogeometry



Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_5) \in \mathbb{Z}_{\geq 0}^5 : \sum_{j=1}^5 j k_j = n - 1\}, \quad K = \sum_{j=1}^5 k_j,$$
$$a(n) = \frac{1}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_5} 15^{k_1} 20^{k_2} 15^{k_3} 6^{k_4} 1^{k_5}.$$
$$a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^5 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-5 coefficient polynomials are embedded in `certificate.payload.json` (source: `case.json#/objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 5. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `case.json#/objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u) / \rho(u)^4$, $\deg_u N = 25$.

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 12×12 ; remainder 5×6 , rank 5, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$15\,A(x) = 14 + 27\,x + A(x)^6$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(3)*T(x)$; $T=x+5*T^2+20*T^3+45*T^4+54*T^5+27*T^6$.
 $a(0), \dots, a(7) = 1, 3, 15, 210, 3510, 65562, 1310901, 27446760$.
 Kernel used throughout: $\rho(u) = u\,(1 - 5\,u^1 - 20\,u^2 - 45\,u^3 - 54\,u^4 - 27\,u^5)$, so $x = \rho(T(x))$.

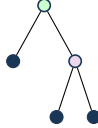
Typogeometry



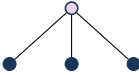
1x3



{1,1}x15



{1,{1,1}}x75



{1,1,1}x60

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 5, Δ_3 : 20, Δ_4 : 45, Δ_5 : 54, Δ_6 : 27.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_5) \in \mathbb{Z}_{\geq 0}^5 : \sum_{j=1}^5 j k_j = n - 1\}, \quad K = \sum_{j=1}^5 k_j,$$

$$a(n) = \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_5} 5^{k_1} 20^{k_2} 45^{k_3} 54^{k_4} 27^{k_5}.$$

$$a(n) = \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^5 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-5 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 5. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n, u) = N(n, u) / \rho(u)^4, \quad \deg_u N = 25.$$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 12×12 ; remainder 5×6 , rank 5, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$31\,A(x) = 30 + 125\,x + A(x)^6$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(5)*T(x)$; $T=x+3*T^2+20*T^3+75*T^4+150*T^5+125*T^6$.
 $a(0),\dots,a(7) = 1, 5, 15, 190, 2550, 38070, 609205, 10199640$.
 Kernel used throughout: $\rho(u) = u\,(1 - 3\,u^1 - 20\,u^2 - 75\,u^3 - 150\,u^4 - 125\,u^5)$, so $x = \rho(T(x))$.

Typogeometry

1x5

{1,1}x15

{1,{1,1}}x45

{1,1,1}x100

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 3, Δ_3 : 20, Δ_4 : 75, Δ_5 : 150, Δ_6 : 125.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1,\dots,k_5) \in \mathbb{Z}_{\geq 0}^5 : \sum_{j=1}^5 j k_j = n-1\}, \quad K = \sum_{j=1}^5 k_j,$$

$$a(n) = \frac{5}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1,\dots,k_5} 3^{k_1} 20^{k_2} 75^{k_3} 150^{k_4} 125^{k_5}.$$

$$a(n) = \frac{5}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^5 P_r(n)a(n+r) = 0$ for $n \geq 1$. The exact degree-5 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 5. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n,u) = N(n,u)/\rho(u)^4, \; \deg_u N = 25.$$

Telescoping identity: $\sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u))$.

Matrix dimensions: 12×12 ; remainder 5×6 , rank 5, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$16\,A(x) = 15 + 27\,x + A(x)^7$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(3)*T(x)$; $T=x+7*T^2+35*T^3+105*T^4+189*T^5+189*T^6+81*T^7$.
 $a(0),\dots,a(7) = 1, 3, 21, 399, 9135, 233709, 6400947, 183585897$.
 Kernel used throughout: $\rho(u) = u\,(1 - 7\,u^1 - 35\,u^2 - 105\,u^3 - 189\,u^4 - 189\,u^5 - 81\,u^6)$, so $x = \rho(T(x))$.

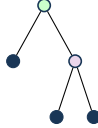
Typogeometry



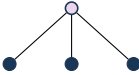
1x3



{1,1}x21



{1,{1,1}}x147



{1,1,1}x105

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 7, Δ_3 : 35, Δ_4 : 105, Δ_5 : 189, Δ_6 : 189, Δ_7 : 81.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1,\dots,k_6) \in \mathbb{Z}_{\geq 0}^6 : \sum_{j=1}^6 j k_j = n-1\}, \quad K = \sum_{j=1}^6 k_j,$$

$$a(n) = \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1,\dots,k_6} 7^{k_1} 35^{k_2} 105^{k_3} 189^{k_4} 189^{k_5} 81^{k_6}.$$

$$a(n) = \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^6 P_r(n)a(n+r) = 0$ for $n \geq 1$. The exact degree-6 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 6. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n,u) = N(n,u)/\rho(u)^5$, $\deg_u N = 36$.
 Telescoping identity: $\sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u))$.
 Matrix dimensions: 14×14 ; remainder 6×7 , rank 6, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$24 A(x) = 23 + 64 x + A(x)^8$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(4)*T(x)$; $T=x+7*T^2+56*T^3+280*T^4+896*T^5+1792*T^6+2048*T^7+1024*T^8$.
 $a(0), \dots, a(7) = 1, 4, 28, 616, 15820, 453208, 13894552, 445970128$.
 Kernel used throughout: $\rho(u) = u \left(1 - 7 u^1 - 56 u^2 - 280 u^3 - 896 u^4 - 1792 u^5 - 2048 u^6 - 1024 u^7\right)$, so $x = \rho(T(x))$.

Typogeometry

1x4

{1,1}x28

{1,{1,1}}x196

{1,1,1}x224

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 7, Δ_3 : 56, Δ_4 : 280, Δ_5 : 896, Δ_6 : 1792, Δ_7 : 2048, Δ_8 : 1024.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_7) \in \mathbb{Z}_{\geq 0}^7 : \sum_{j=1}^7 j k_j = n - 1\}, \quad K = \sum_{j=1}^7 k_j,$$

$$a(n) = \frac{4}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_7} 7^{k_1} 56^{k_2} 280^{k_3} 896^{k_4} 1792^{k_5} 2048^{k_6} 1024^{k_7}.$$

$$a(n) = \frac{4}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^7 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-7 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 7. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u) / \rho(u)^6$, $\deg_u N = 49$.
 Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.
 Matrix dimensions: 16×16 ; remainder 7×8 , rank 7, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$25 A(x) = 24 + 64 x + A(x)^9$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(4)*T(x)$; $T=x+9*T^2+84*T^3+504*T^4+2016*T^5+5376*T^6+9216*T^7+9216*T^8+4096*T^9$.
 $a(0), \dots, a(7) = 1, 4, 36, 984, 31716, 1140552, 43895208, 1768717872$.
 Kernel used throughout: $\rho(u) = u(1 - 9u^1 - 84u^2 - 504u^3 - 2016u^4 - 5376u^5 - 9216u^6 - 9216u^7 - 4096u^8)$, so $x = \rho(T(x))$.

Typogeometry

1x4

{1,1}x36

{1,{1,1}}x324

{1,1,1}x336

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 9, Δ_3 : 84, Δ_4 : 504, Δ_5 : 2016, Δ_6 : 5376, Δ_7 : 9216, Δ_8 : 9216, Δ_9 : 4096.

Coefficient and contour extraction

$$\begin{aligned}
 \mathcal{K}_n &= \{(k_1, \dots, k_8) \in \mathbb{Z}_{\geq 0}^8 : \sum_{j=1}^8 j k_j = n - 1\}, \quad K = \sum_{j=1}^8 k_j, \\
 a(n) &= \frac{4}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1, \dots, k_8} 9^{k_1} 84^{k_2} 504^{k_3} 2016^{k_4} 5376^{k_5} 9216^{k_6} 9216^{k_7} 4096^{k_8}. \\
 a(n) &= \frac{4}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.
 \end{aligned}$$

Recurrence

$\sum_{r=0}^8 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-8 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 8. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

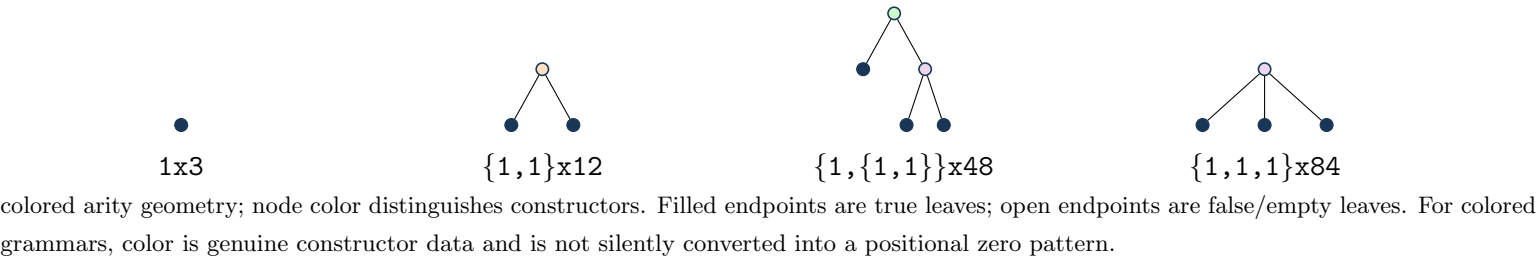
Certificate.

$R(n, u) = N(n, u) / \rho(u)^7$, $\deg_u N = 64$.
 Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.
 Matrix dimensions: 18×18 ; remainder 8×9 , rank 8, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$36 A(x) = 35 + 81 x + A(x)^9$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(3)*T(x)$; $T=x+4*T^2+28*T^3+126*T^4+378*T^5+756*T^6+972*T^7+729*T^8+243*T^9$.
 $a(0), \dots, a(7) = 1, 3, 12, 180, 3018, 56238, 1121484, 23406804$.
 Kernel used throughout: $\rho(u) = u \left(1 - 4 u^1 - 28 u^2 - 126 u^3 - 378 u^4 - 756 u^5 - 972 u^6 - 729 u^7 - 243 u^8\right)$, so $x = \rho(T(x))$.

Typogeometry



Typogeometric constructor codes and multiplicities: Δ_2 : 4, Δ_3 : 28, Δ_4 : 126, Δ_5 : 378, Δ_6 : 756, Δ_7 : 972, Δ_8 : 729, Δ_9 : 243.

Coefficient and contour extraction

$$\begin{aligned}
 \mathcal{K}_n &= \{(k_1, \dots, k_8) \in \mathbb{Z}_{\geq 0}^8 : \sum_{j=1}^8 j k_j = n - 1\}, \quad K = \sum_{j=1}^8 k_j, \\
 a(n) &= \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_8} 4^{k_1} 28^{k_2} 126^{k_3} 378^{k_4} 756^{k_5} 972^{k_6} 729^{k_7} 243^{k_8}. \\
 a(n) &= \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.
 \end{aligned}$$

Recurrence

$\sum_{r=0}^8 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-8 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 8. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

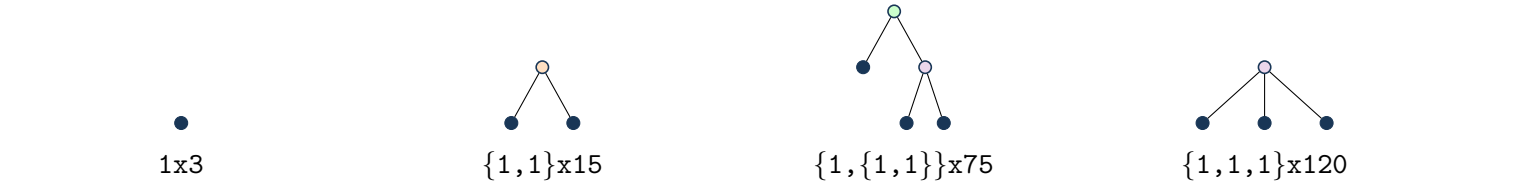
Certificate.

$R(n, u) = N(n, u) / \rho(u)^7$, $\deg_u N = 64$.
 Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.
 Matrix dimensions: 18×18 ; remainder 8×9 , rank 8, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$37 A(x) = 36 + 81 x + A(x)^{10}$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(3)*T(x)$; T-polynomial coefficients are the Delta multiplicities below.
 $a(0), \dots, a(7) = 1, 3, 15, 270, 5505, 124818, 3028200, 76896180$.
 Kernel used throughout: $\rho(u) = u \left(1 - 5 u^1 - 40 u^2 - 210 u^3 - 756 u^4 - 1890 u^5 - 3240 u^6 - 3645 u^7 - 2430 u^8 - 729 u^9\right)$, so $x = \rho(T(x))$.

Typogeometry



colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_{10} : 729, Δ_2 : 5, Δ_3 : 40, Δ_4 : 210, Δ_5 : 756, Δ_6 : 1890, Δ_7 : 3240, Δ_8 : 3645, Δ_9 : 2430.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_9) \in \mathbb{Z}_{\geq 0}^9 : \sum_{j=1}^9 j k_j = n - 1\}, \quad K = \sum_{j=1}^9 k_j,$$

$$a(n) = \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_9} 5^{k_1} 40^{k_2} 210^{k_3} 756^{k_4} 1890^{k_5} 3240^{k_6} 3645^{k_7} 2430^{k_8} 729^{k_9}.$$

$$a(n) = \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^9 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-9 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 9. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n, u) = N(n, u) / \rho(u)^8, \quad \deg_u N = 81.$$

$$\text{Telescoping identity: } \sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u)).$$

Matrix dimensions: 20×20 ; remainder 9×10 , rank 9, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$(4-1\,x)\,A(x)=3+A(x)^3$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+1*x*T+3*T^2+1*T^3$.
 $a(0),\ldots,a(7)=1,1,4,29,263,29088,331749$.
 Kernel used throughout: $\rho(u)=u\frac{1-3\,u^1-1\,u^2}{1+1\,u}$, so $x=\rho(T(x))$.

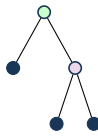
Typogeometry



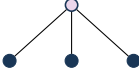
1x1



{1,1}x4



{1,{1,1}}x16



{1,1,1}x1

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 3, Δ_2^{marked} : 1, Δ_3 : 1.

Coefficient and contour extraction

$$\mathcal{K}_{n,h}=\{(k_1,\ldots,k_2)\in\mathbb{Z}_{\geq 0}^2:h+\sum_{j=1}^2jk_j=n-1\},\quad K=\sum_{j=1}^2k_j,$$

$$a(n)=\frac{1}{n}\sum_{h=0}^{n-1}\binom{n}{h}\sum_{\boldsymbol{k}\in\mathcal{K}_{n,h}}\binom{n+K-1}{K}\binom{K}{k_1,\ldots,k_2}3^{k_1}1^{k_2}.$$

$$a(n)=\frac{1}{2\pi i n}\oint_\gamma\frac{du}{\rho(u)^n},\quad n\geq 1.$$

Recurrence

$$\sum_{r=0}^4P_r(n)a(n+r)=0,\,n\geq 1,\,\text{with}$$

$$\begin{aligned}P_0(n)&=4\,n^3+2\,n^2-2\,n\\P_1(n)&=-64\,n^3-168\,n^2-136\,n-32\\P_2(n)&=384\,n^3+1824\,n^2+2848\,n+1472\\P_3(n)&=-781\,n^3-5825\,n^2-14286\,n-11520\\P_4(n)&=52\,n^3+468\,n^2+1352\,n+1248\end{aligned}$$

Differential equation

$$\begin{aligned}(0)\,A(x)&+(4\,x^5-32\,x^4+64\,x^3)\,A'(x)\\&+(14\,x^6-168\,x^5+672\,x^4-1139\,x^3)\,A^{(2)}(x)+(4\,x^7-64\,x^6+384\,x^5-781\,x^4+52\,x^3)\,A^{(3)}(x)=0.\end{aligned}$$

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n,u,x)=N/D\text{ (exact 246-character numerator in embedded payload).}$$

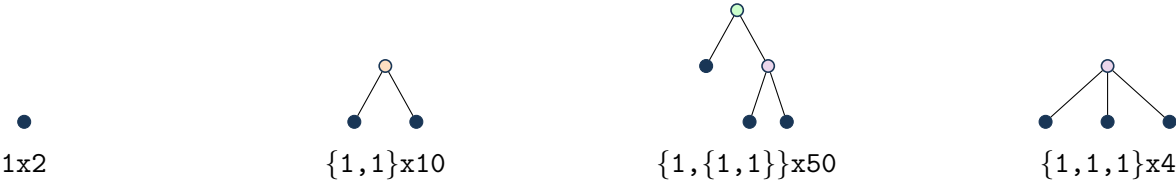
$$\text{Telescoping identity: }\sum_rP_r(n)H_{n+r}(u)=\partial_u(R(n,u)H_n(u)).$$

Matrix dimensions: 6×6; remainder 2×3, rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$(5 - 4x) A(x) = 4 + A(x)^3$, with the origin-normalized solution.
Normalization/grammar: $A(x)=1+(2)*T(x)$; $T=x+2*x*T+3*T^2+2*T^3$.
 $a(0), \dots, a(7) = 1, 2, 10, 84, 882, 10380, 130916, 1729960$.
Kernel used throughout: $\rho(u) = u \frac{1 - 3u^1 - 2u^2}{1 + 2u}$, so $x = \rho(T(x))$.

Typogeometry



colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.
Typogeometric constructor codes and multiplicities: $\Delta_2: 3$, $\Delta_2^{\text{marked}}: 2$, $\Delta_3: 2$.

Coefficient and contour extraction

$$\mathcal{K}_{n,h} = \{(k_1, \dots, k_2) \in \mathbb{Z}_{\geq 0}^2 : h + \sum_{j=1}^2 j k_j = n - 1\}, \quad K = \sum_{j=1}^2 k_j,$$
$$a(n) = \frac{2}{n} \sum_{h=0}^{n-1} \binom{n}{h} \sum_{\mathbf{k} \in \mathcal{K}_{n,h}} \binom{n+K-1}{K} \binom{K}{k_1, \dots, k_2} 3^{k_1} 1^{k_2}.$$

$$a(n) = \frac{2}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^4 P_r(n) a(n+r) = 0$, $n \geq 1$, with

$$\begin{aligned} P_0(n) &= 256 n^3 + 128 n^2 - 128 n \\ P_1(n) &= -1280 n^3 - 3360 n^2 - 2720 n - 640 \\ P_2(n) &= 2400 n^3 + 11400 n^2 + 17800 n + 9200 \\ P_3(n) &= -1568 n^3 - 11590 n^2 - 28158 n - 22500 \\ P_4(n) &= 85 n^3 + 765 n^2 + 2210 n + 2040 \end{aligned}$$

Differential equation

The scalar linear ODE has order 3. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `runs/A244627-direct-x-pilot/case.json#ode_for_A`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u, x) = N/D$ (exact 275-character numerator in embedded payload).

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 6×6 ; remainder 2×3 , rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Definition and first terms

$(5 - 1\,x)\,A(x) = 4 + A(x)^4$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+1*x*T+6*T^2+4*T^3+1*T^4$.
 $a(0),\dots,a(7) = 1, 1, 7, 95, 1614, 30718, 626434, 13383650$.
 Kernel used throughout: $\rho(u) = u\frac{1-6\,u^1-4\,u^2-1\,u^3}{1+1\,u}$, so $x = \rho(T(x))$.

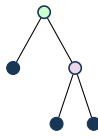
Typogeometry



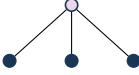
1x1



{1,1}x7



{1,{1,1}}x49



{1,1,1}x4

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 6, Δ_2^{marked} : 1, Δ_3 : 4, Δ_4 : 1.

Coefficient and contour extraction

$$\mathcal{K}_{n,h} = \{(k_1,\dots,k_3) \in \mathbb{Z}_{\geq 0}^3 : h + \sum_{j=1}^3 j k_j = n-1\}, \quad K = \sum_{j=1}^3 k_j,$$

$$a(n) = \frac{1}{n} \sum_{h=0}^{n-1} \binom{n}{h} \sum_{\mathbf{k} \in \mathcal{K}_{n,h}} \binom{n+K-1}{K} \binom{K}{k_1,\dots,k_3} 6^{k_1} 4^{k_2} 1^{k_3}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^4 P_r(n) a(n+r) = 0, \, n \geq 0, \text{ with}$$

$$P_0(n) = 3\,(n+1)\,(3\,n-1)\,(3\,n+7)$$

$$P_1(n) = -15\,(2\,n+3)\,(18\,n^2+54\,n+29)$$

$$P_2(n) = 75\,(n+2)\,(54\,n^2+216\,n+209)$$

$$P_3(n) = -6750\,(n+2)\,(n+3)\,(2\,n+5)$$

$$P_4(n) = 491\,(n+2)\,(n+3)\,(n+4)$$

Differential equation

The scalar linear ODE has order 3. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/attached_order4_certificate.json#/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$C(n,t) = \frac{(1+t)P(n,t)}{n(n+1)t^3Q(t)^3}.$$

Telescoping identity: $\sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u))$.

Matrix dimensions: 8×8 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.